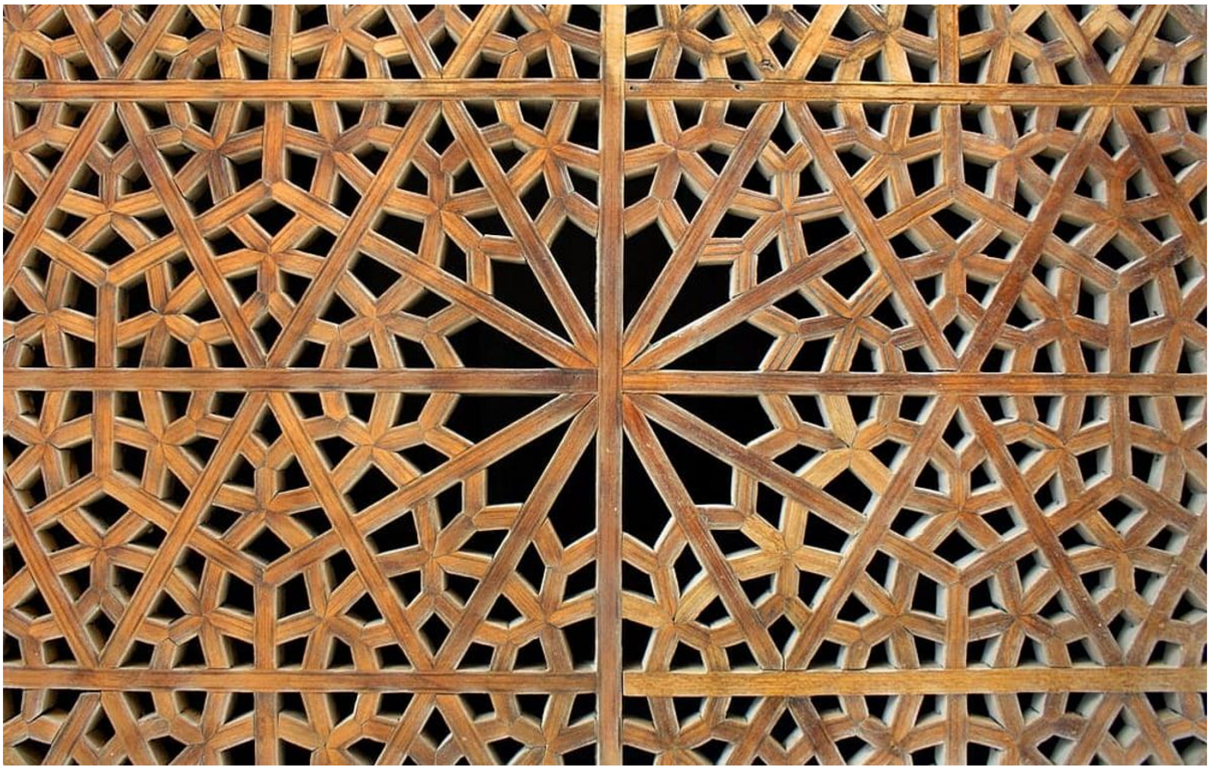
CIE 2 – 1RUA24CSE0418, Shaikh Abrar Ahmed

Task 1: Craft Documentation

Selected Craft: *Pinjra Kari* (Kashmiri Interlocking Woodwork)

1. Overview *Pinjra Kari* is a traditional craft from Kashmir crafting intricate, geometric wooden latticework. It is used primarily for windows, doors, and screens. The defining characteristic of this craft is that the wooden slats are intricately joined together using precise interlocking joints without the use of a nail or glue.




Outlook
traveller

2. Tools

- **Aari (Fine-toothed Saw):** A specialized hand saw used to make precise, narrow cuts for the tiny wooden laths.
- **Randa (Hand Plane):** Used for smoothing the raw timber into thin strips before they are cut into interlocking components.
- **Teshna / Nihan (Chisels):** Fine chisels used to carve out the tiny half-lap joints, mortises, and tenons.
- **Khat-kash (Marking Gauge & Compass):** Because the craft relies on strict geometry, compasses and traditional marking tools are required

Khat-Kash Marking gauge

Randa (Hand Plane)

3. Materials

- **Deodar (Himalayan Cedar):** The standard for Pinjra Kari. It has a very straight grain, doesn't warp easily, and is naturally packed with oils that make it highly resistant to termites and harsh weather.
- **Kail (Blue Pine) or Walnut:** Sometimes used for interior screens or furniture pieces.

Deodar Wood

Blue pine wood

4. Processes

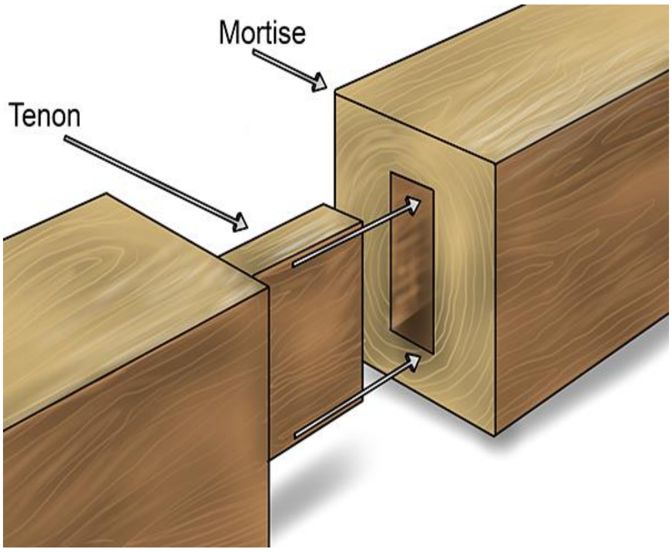
1. **Rasm (Geometric Layout):** The craftsman draws a complex geometric grid on paper or directly on a template. The designs are mathematically rigorous, often based on 6-, 8-, or 12-pointed stars.
2. **Dimensioning:** The wood is planed and ripped into perfectly uniform, thin battens (laths).
3. **Tukrai (Cutting the Joints):** The craftsman cuts precise slots, half-laps, and tiny tenons into the laths at exact angles corresponding to the geometric grid.
4. **Jodna (Assembly):** The pieces are woven and tapped together. The magic of Pinjra Kari is that the friction and the opposing tension of the geometric angles lock the entire lattice tightly in place.

5. Cultural Context: Introduced to Kashmir from Persia around the 14th century, Pinjra Kari became a part of local architecture, from houseboats to shrines. Functionally, these screens provided privacy (allowing those inside to see out without being seen) and ventilation while acting as a barrier to winter snow. Culturally, the repetitive, interlocking geometric patterns are rooted in Islamic art, symbolizing the infinite nature of the universe and unity.

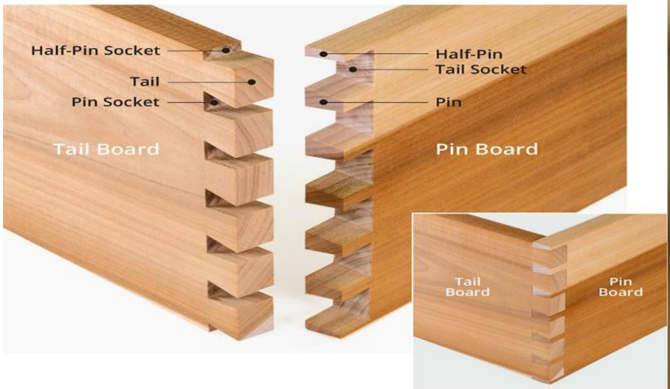
Task 2: Joinery Sheet

Wood Joineries:

1. **Mortise and Tenon:** * *Description:* A projecting pin (tenon) on one piece fits into a corresponding hole (mortise) on another.
    - *Function:* Extremely strong structural joint used for heavy frames, table legs, and door construction.

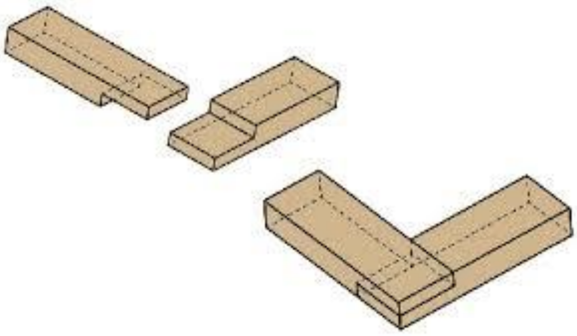


2. **Through Dovetail:**
    - *Description:* Flaring, wedge-shaped interlocking "tails" and "pins" that resemble a dove's tail.
    - *Function:* Highly resistant to being pulled apart (tensile strength); heavily used in high-quality box and drawer construction.

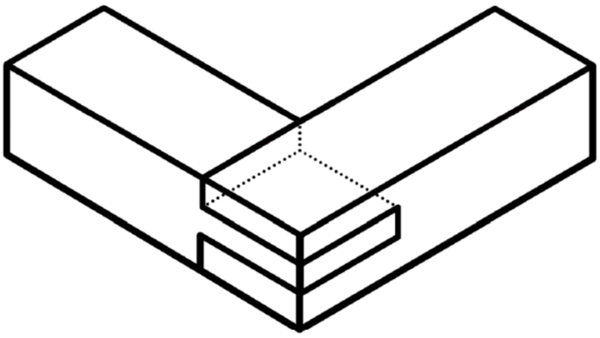


3. **Half-Lap Joint:**

- *Description:* Half the thickness of each of two intersecting timbers is removed so they sit flush with one another.
- *Function:* Used where two members cross in the same plane, common in lattice work (like Pinjra Kari) and framing.

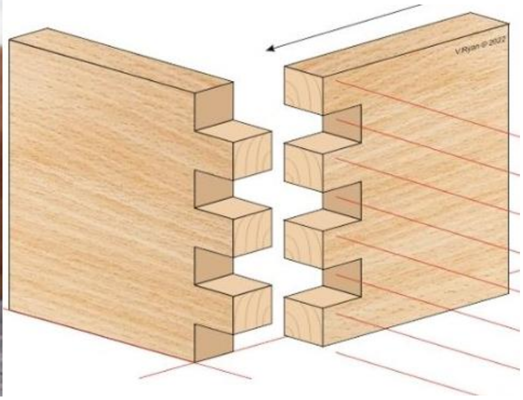


4. **Bridle Joint:**
   - *Description:* Similar to a mortise and tenon, but the mortise is cut as an open slot at the end of the wood.
   - *Function:* Excellent for corner joints in frames where a large gluing surface is needed.



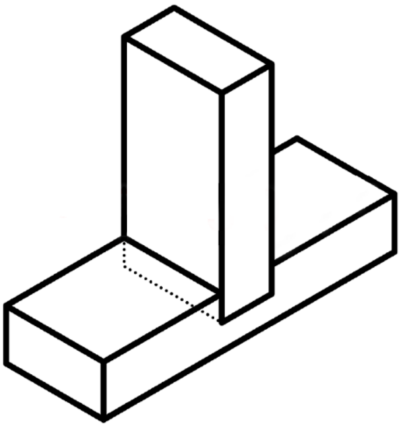
5. **Finger (Box) Joint:**

   - *Description:* A series of rectangular, interlocking cuts on the ends of two pieces.
   - *Function:* Greatly increases the glue surface area for corner joining; ideal for boxes and cabinetry.

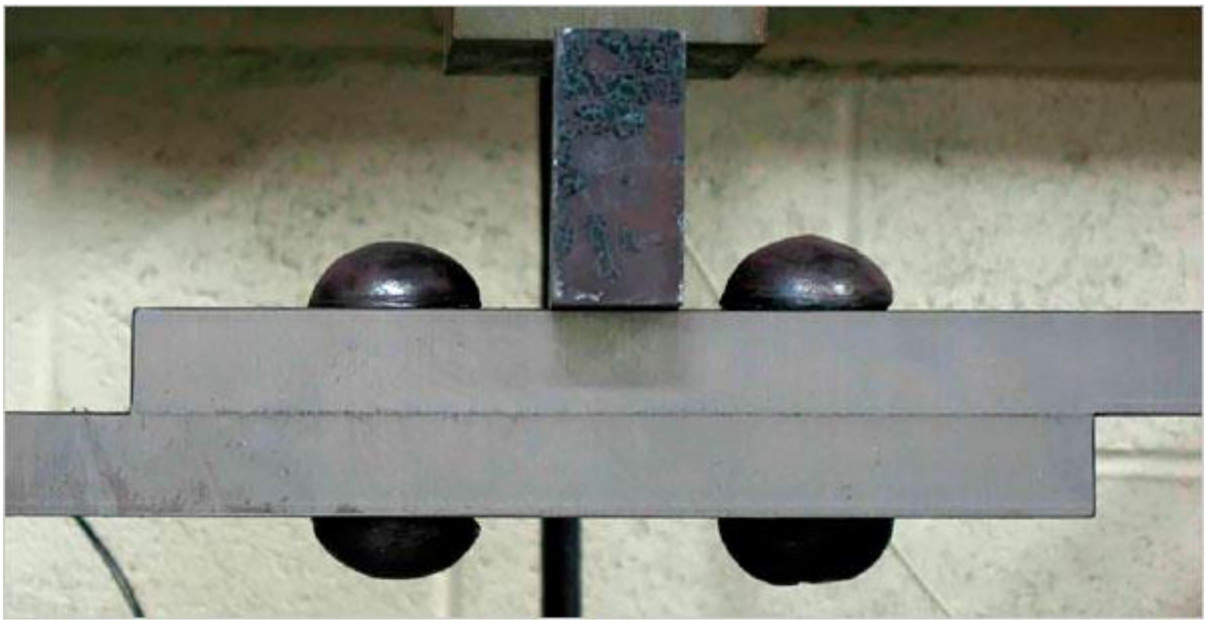
6. **Dado Joint:**

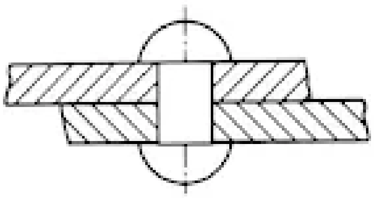
   - *Description:* A three-sided trench cut across the grain of one piece to receive the end of another.
   - *Function:* The standard joint for building shelvin g units and cabinets, providing excellent shear support.

Metal Joineries:

7. **Riveted Lap Joint:**
    - *Description:* A permanent mechanical fastener consisting of a smooth cylindrical shaft with a head, where the tail is deformed (bucked) to hold plates together.
    - *Function:* Used for permanent structural connections in sheet metal, aircraft, and traditional steel structures (like bridges).

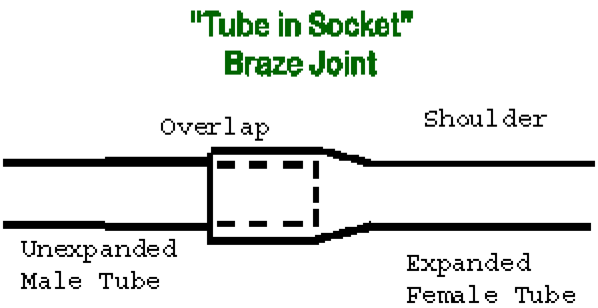




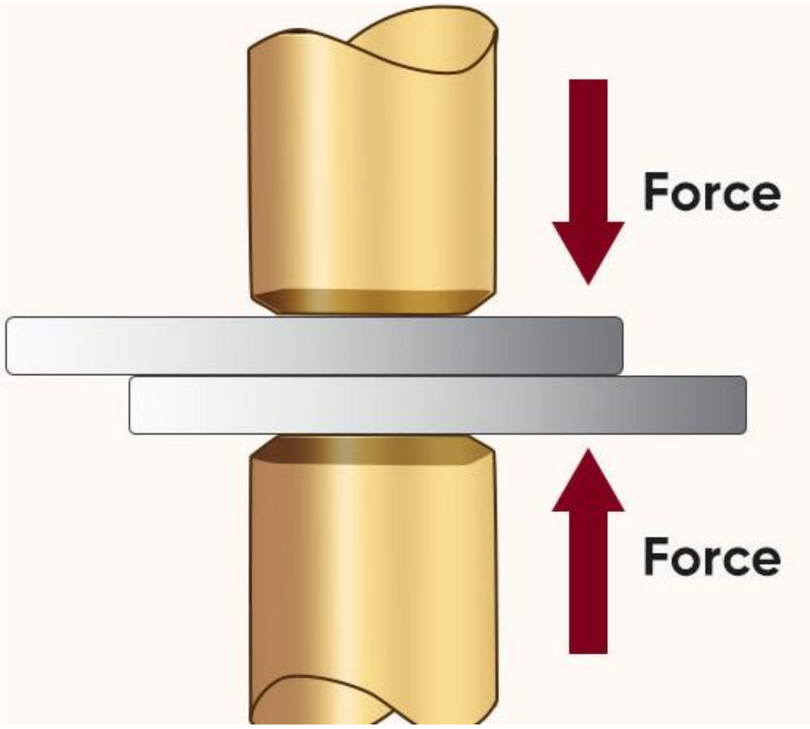
Lap Joint

8. **Brazed Joint:**

   - *Description:* Joining metals by heating them and melting a filler metal (like brass) into the joint via capillary action, without melting the base metal.
   - *Function:* Excellent for joining dissimilar metals, delicate frames, and pipe fittings where welding might warp the material.

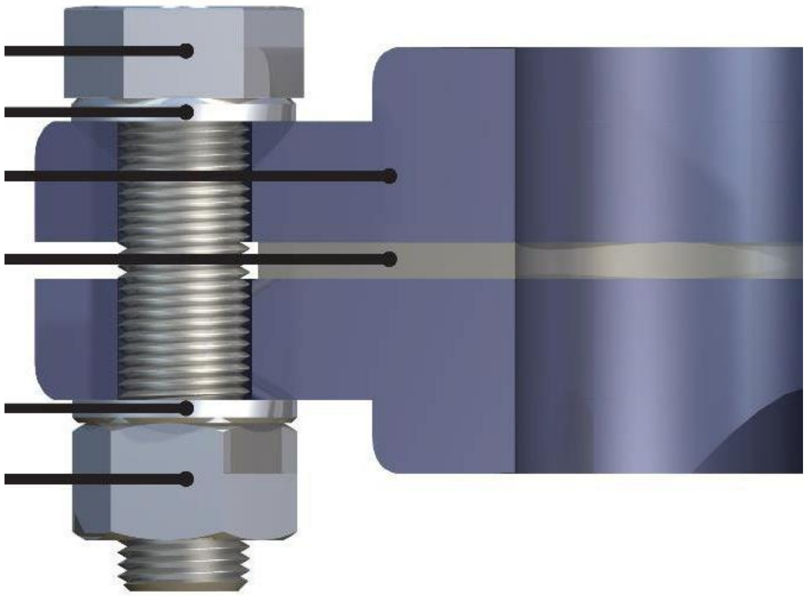


9. **Spot Weld:**
    - *Description:* Fusing overlapping metal sheets by applying pressure and heat from an electric current to an isolated spot.
    - *Function:* Highly efficient for quick, permanent bonding of thin sheet metals (widely used in automotive frames).

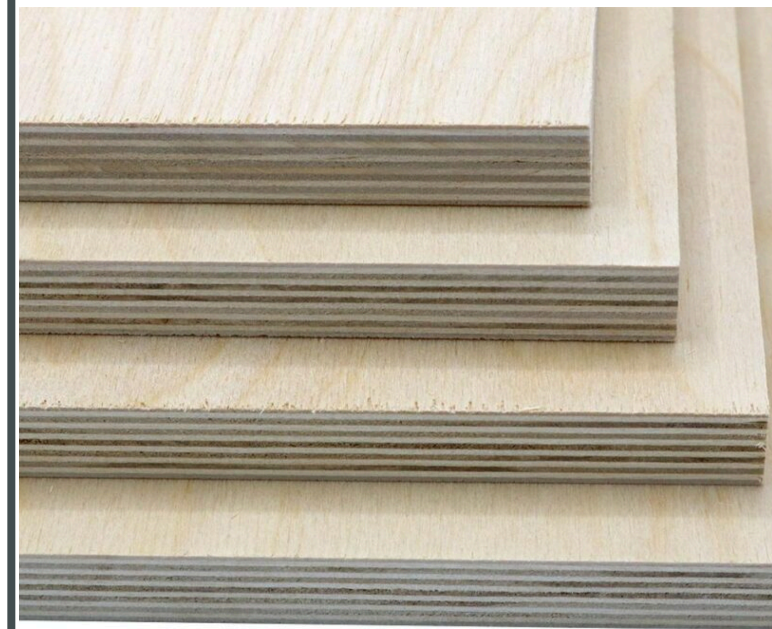


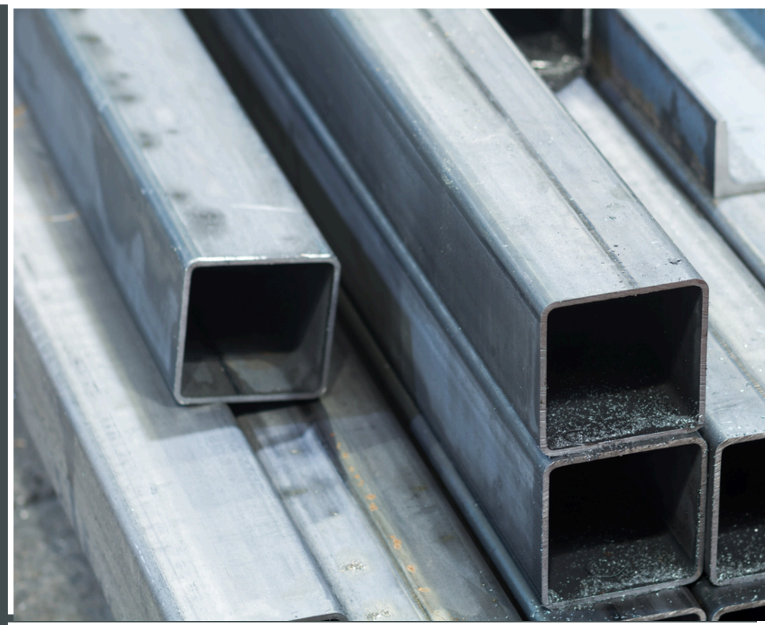
10. **Bolted Flange:**

   - *Description:* Two pieces of metal with flared rims (flanges) connected by bolts compressing a gasket between them.
   - *Function:* Used for strong structural connections that need to be disassembled later, such as in heavy steel beams or plumbing pipes.



Task 3: Material Study - 1RUA24CSE0418

MATERIAL	KEY PROPERTIES	STRUCTURAL BEHAVIOUR	JOINERY COMPATIBILITY	VISUAL
1. Deodar Wood (Softwood)	Lightweight, straight-grained, naturally weather/ rot resistant.	High stability; doesn't warp easily under tension. Moderate compressive strength.	Excellent for intricate tension joints (like Pinjra Kari), half-laps, and glued joints.	
2. White Oak (Hardwood)	Dense, heavy, high wear resistance, beautiful prominent grain.	Very high tensile and compressive strength. Brittle if bent abruptly without steam.	Ideal for heavy-duty mortise and tenon, dovetails, and peg-and-hole joinery.	
3. Baltic Birch Plywood	Uniform thickness, void-free core, isotropic (strength in all directions).	Highly resistant to warping and twisting; excellent sheer strength. Edges can splinter.	Best with dadoes, rabbets, and mechanical fasteners (screws). Poor for traditional dovetails.	

MATERIAL	KEY PROPERTIES	STRUCTURAL BEHAVIOUR	JOINERY COMPATIBILITY	VISUAL
4. Mild Steel	Ductile, malleable, magnetic, prone to oxidation (rust).	High tensile and compressive strength. Deflects under load before failing (yielding).	Best suited for welding (MIG/TIG), bolting, and riveting.	
5. Brass (Alloy)	Corrosion-resistant, low friction, aesthetically warm, easy to machine.	Softer than steel; moderate structural strength. Good acoustic properties.	Excellent for brazing, soldering, and tapped (threaded) mechanical joints.	
6. Aluminum (6061)	High strength-to-weight ratio, non-magnetic, highly corrosion-resistant.	Stiff but lightweight. Can suffer from metal fatigue under repeated cyclical stress.	Riveting, bolting, and specialized TIG welding. Hard to braze or solder.